

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	25 SEPTEMBER 2026
Team ID	SWTID-2026-4636
Project Name	AI BLOGNEXT
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

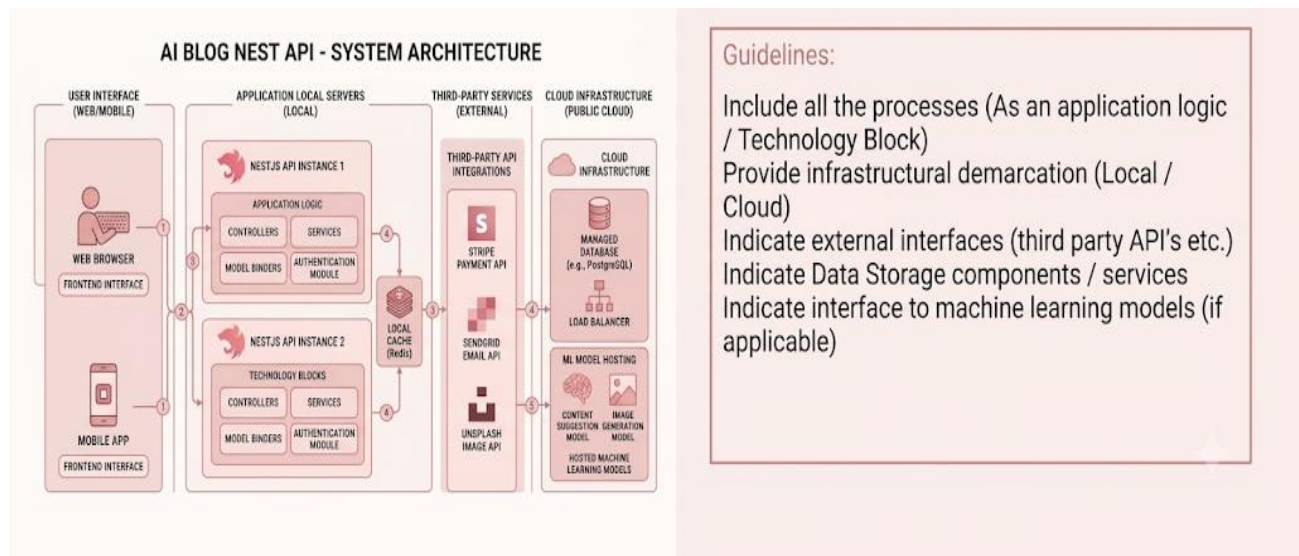


Table-1: Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Provides an interface through which bloggers and content creators can register, log in, create blogs, manage posts, and use AI content-generation features.	HTML, CSS, JavaScript / React.js
2	Application Logic-1	Handles user registration, login, authentication, blog creation, updating, viewing, and deletion.	Node.js, Express.js
3	Application Logic-2	Processes AI-based requests and generates blog content based on the topic or prompt provided by the user.	Python / AI API

4	Application Logic-3	Handles authentication, validation, API routing, error handling, and communication between the frontend and database.	Express.js, REST API
5	Database	Stores user accounts, blog posts, categories, generated content, and other application information.	MongoDB
6	Cloud Database	Provides optional cloud-based database storage for accessing application data from different environments.	MongoDB Atlas
7	File Storage	Stores optional blog-related files such as images, documents, or other uploaded content.	Local File System / Cloud Storage
8	External API-1	Provides AI-powered content generation based on user-provided topics and prompts.	Generative AI API
9	External API-2	Can be used to provide additional services such as image generation or other content-related features.	Image/Content API
10	Machine Learning Model	Supports intelligent content generation, content improvement, summarization, and other AI-assisted blogging functions.	Generative AI / NLP Model
11	Infrastructure (Server / Cloud)	Hosts and runs the backend API and database services. The application can initially be developed locally and later deployed to a cloud platform.	Node.js Server / Cloud Hosting

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Uses open-source technologies for developing the frontend, backend, API services, and database integration.	Node.js, Express.js, React.js, MongoDB
2	Security Implementations	Protects user accounts and blog data using password hashing, JWT authentication, input validation, authorization, and protected API routes.	JWT, bcrypt, HTTPS, Input Validation
3	Scalable Architecture	Uses a modular REST API architecture where frontend, backend, database, and AI services can be developed and expanded independently.	Node.js, Express.js, MongoDB REST API
4	Availability	The application can be deployed on a reliable cloud server and database service to allow users to access the system whenever the service is available.	Cloud Hosting, MongoDB Atlas

5	Performance	Uses efficient REST APIs, database queries, validation, and modular backend processing to provide quick responses for blog management operations. AI requests are processed separately to avoid unnecessary load on regular API operations.	Node.js, Express.js, MongoDB
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Technology Stack Summary

Technology	Details
Frontend	HTML, CSS, JavaScript / React.js
Backend	Node.js + Express.js
Database	SQLite
Authentication	JWT + bcrypt
API	RESTful API
AI	Generative AI API / NLP Model
API Testing	Postman
Development Tool	Visual Studio Code
Version Control	Git / GitHub
Deployment	Local Server / Cloud Hosting
Data Format	JSON
Architecture	REST-based Client-Server Architecture

Technology Description

- **Node.js:** Used as the server-side JavaScript runtime.
- **Express.js:** Used to create RESTful API routes and handle HTTP requests.
- **SQLite:** Used as the lightweight relational database for storing users and blog information.
- **JWT:** Used for secure user authentication and protected API access.
- **bcrypt:** Used to securely hash user passwords.
- **Generative AI API:** Used for AI-powered blog content generation and summarization.
- **Postman:** Used to test API endpoints, authentication, request validation and responses.
- **VS Code:** Used as the primary development environment.
- **JSON:** Used for communication between the client and REST API.